AP CALCULUS AB

Unit 1 • Lesson 1.10 notes

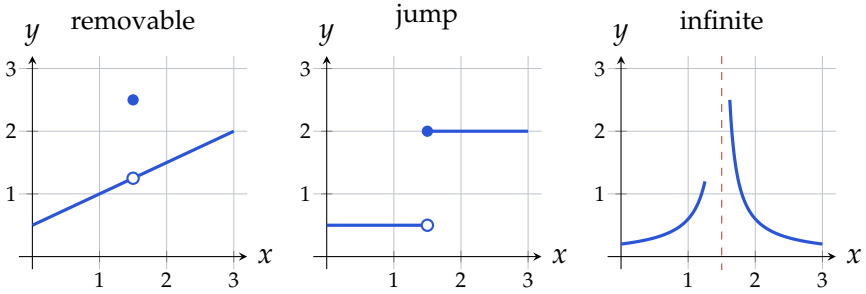
Lesson 1.10 • Exploring Types of Discontinuities

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Classify discontinuities as removable, jump, or infinite.
- Justify the classification with limits.

Vocabulary. continuous, discontinuity, removable, jump, infinite.



Key idea

- **Removable**: $\lim_{x\to c} f(x)$ exists, but $f(c)$ is undefined or different from it.
- **Jump**: both one-sided limits exist (finite) and are different.
- **Infinite**: at least one one-sided limit is $+\infty$ or $-\infty$.

Example 1 • Formulas

Solution. $\dfrac{x^2-9}{x-3}$ at $x=3$: limit 6, undefined value: removable. $\dfrac{|x|}{x}$ at 0: limits -1 and 1: jump. $\dfrac{1}{x^2}$ at 0: limit $+\infty$: infinite.

Example 2 • Piecewise

$$f(x)=\begin{cases} x+1, & x<1 \\ 3, & x=1 \\ 3-x, & x>1. \end{cases}$$

Solution. $\lim_{x\to 1^-} f(x) = 2$ and $\lim_{x\to 1^+} f(x) = 2$, so the limit is 2, but $f(1) = 3$: removable discontinuity, fixed by redefining $f(1) = 2$.

Common mistakes

- Calling every point where f is undefined an asymptote; check whether the limit is finite.
- Classifying a jump as removable because both one-sided limits exist.

Quick check

Classify at the given point: 1. $\frac{x+2}{x^2-4}$ at $x=-2$ 2. same function at $x=2$ 3. $\begin{cases} x^2, & x\le 0 \\ x+1, & x>0 \end{cases}$ at 0.

Answers. 1. Removable (limit $-\frac{1}{4}$). 2. Infinite. 3. Jump.